Notes: Blickle, Parlatore, and Saunders (JF, 2026)

Specialization in Banking

Contents

1 Big picture **3**

2 Data and measurement (key objects) **3**

2.1 Core supervisory loan panel . . . 3

2.2 Auxiliary data and extra measurements . . . 4

2.3 Key constructed objects . . . 5

2.4 Timing and empirical applications . . . 5

3 Models and empirical specifications (all equations + interpretation) **5**

3.1 Economic mechanism and hypotheses . . . 5

3.2 Measuring specialization . . . 6

3.3 Baseline loan-performance regression . . . 6

3.4 Why ratings and rates matter for interpretation . . . 7

3.5 Alternative performance specifications . . . 7

3.6 Loan-terms regressions . . . 7

3.7 Bank-level and deposit-shock exercises . . . 8

4 Identification and instruments (what makes the estimates credible) **8**

4.1 What makes the empirical question hard . . . 8

4.2 What the paper does to make the results credible . . . 8

4.3 How this relates to the literature . . . 9

5 Tables and figures **9**

5.1 Table I: Summary statistics of key variables . . . 9

5.2 Figure 1: Excess specialization over time . . . 10

5.3 Table II: Summary statistics of specialization . . . 11

5.4 Figure 2: Distribution of specialization and bank size . . . 11

5.5 Figure 3: Raw loan-term and performance differences 12
5.6 Table III: Specialization and loan performance 12
5.7 Table IV: Alternative concentration measures and the horse race 13
5.8 Table V: Different specialization definitions 13
5.9 Table VI: Specialization and loan terms 13
5.10 Aggregate results and extensions from Sections VII–VIII 16

6 Short summary **16**

7 Conclusion **17**
7.1 Core conclusion (what the paper establishes) 17
7.2 Economic intuition 17
7.3 Takeaway 17

1 Big picture

- **Question.** Do large U.S. banks **specialize** their commercial and industrial (C&I) loan portfolios in a few industries, and if they do, is that specialization consistent with **industry-specific information** that improves loan performance?
- **Setting.** The paper studies detailed supervisory loan-level data for the largest stress-tested U.S. banks, then supplements these data with syndicated-loan data, bank balance-sheet data, branch deposit data, and Compustat information.
- **Core challenge.** A specialized bank may look different from other banks for many reasons that have nothing to do with information: balance-sheet constraints, monopoly-like industry capture, regional concentration, repeated bank-firm relationships, or simple selection of safer borrowers. The paper therefore has to distinguish **active, information-based specialization** from **passive concentration**.
- **Main descriptive result.** Even very large banks are underdiversified in a systematic way. The average bank directs about **9 percentage points more** of its C&I portfolio to its favorite two-digit industry than a fully diversified bank would, and this preferred-industry pattern is persistent over time.
- **Main loan-performance result.** Loans made in a bank's specialized industry are less likely to become nonperforming. This pattern survives rich controls for loan characteristics, internal risk ratings, relationship lending, regional concentration, and saturated bank-time and industry-time fixed effects.
- **Main loan-terms result.** Specialized lenders offer terms that are more attractive to borrowers they want to keep: larger loans, lower rates, and longer maturity, especially when the loan is unsecured or the borrower has outside funding options.
- **Main bank-level result.** Specialization is associated with more stable performance and fewer charge-offs, but also somewhat lower profitability. Banks appear to trade off return for stability and concentrate more on their preferred industries when Tier 1 capital is relatively low.
- **Broader implication.** Banks are not perfectly fungible. When deposits move across the banking system—for example during COVID or after stress in specific banks—the industries favored by the receiving banks can get disproportionately more credit.
- **Why this paper matters.** The paper adds an industry-specialization channel to the banking literature. It connects loan portfolio composition to information production, loan performance, borrower opacity, bank stability, and the real effects of deposit reallocation.

2 Data and measurement (key objects)

2.1 Core supervisory loan panel

- **Main source.** FR Y-14 Q, subdatabase H.1, maintained by the Federal Reserve for stress testing.
- **Coverage.** Quarterly C&I loan holdings of stress-tested banks, with loans reported when commitment exceeds \$1 million.

- **Sample window.** 2012Q2–2023Q2.
- **Banks.** 40 large U.S. banks.
- **Sample used in the main tests.** About **297,000 term loans** and more than **3 million loan-quarter observations** after cleaning.
- **Market coverage.** The data cover more than **75% of U.S. C&I lending by dollar volume** during the sample period.
- **Observed loan variables.** Loan size, interest rate, maturity, collateral status and collateral type, loan type, loan purpose, internal risk rating, arrears/default/nonaccrual flags, borrower name, borrower location, and borrower industry.
- **Industry coding.** Two-digit and four-digit NAICS codes in Y-14.
- **Main outcome.** A loan is **nonperforming** if it ever falls at least 90 days into arrears, becomes nonaccrual/default, or carries negative remaining maturity because payments were not completed by maturity.

2.2 Auxiliary data and extra measurements

- **SNC data.** The Shared National Credit registry is used for smaller lenders and for a longer period, helping the paper distinguish specialization from simple balance-sheet constraints.
- **Y-9C data.** Quarterly bank-level balance-sheet and income data, used to study profitability, risk, Tier 1 capital, and aggregate performance.
- **FR2900 weekly deposits.** Used to study the allocation of exogenous deposit inflows during COVID and related episodes.
- **Branch deposit data.** Used to construct zip-code-level deposit concentration and excess regional lending measures.
- **Compustat links.** Used in extensions on borrower opacity, SME lending, and real effects on firm growth and earnings.

The main summary statistics for the Y-14 term-loan sample are:

- average log committed amount: 8.42,
- average interest rate: 3.80%,
- average remaining maturity: 24.5 quarters,
- unsecured share: 18%,
- ever-nonperforming share: 4%,
- average bank share of a borrower's industry ("industry capture"): 9%,
- average number of bank-borrower interactions: about 9.3.

2.3 Key constructed objects

- **Portfolio share in an industry.** The share of a bank's C&I portfolio invested in a given industry or sector.
- **Excess specialization.** The degree to which the bank overinvests in an industry relative to a fully diversified portfolio weighted by industry size.
- **Relative specialization.** The ratio of the bank's portfolio share in an industry to that industry's share in aggregate C&I lending.
- **Favorite industry.** The industry in which a bank is most overinvested.
- **Industry capture.** The share of all Y-14 lending to an industry accounted for by a single bank.
- **Regional concentration.** The share of a bank's lending in a zip code or state.
- **Relationship lending.** The number of observed interactions between a bank and a borrower.

2.4 Timing and empirical applications

- **Timing.** Specialization is measured at loan origination, and the dependent variable tracks whether the loan performs badly at any point in the future.
- **Internal ratings at origination.** These are useful because they are standardized for supervisory purposes. Conditioning on them lets the paper ask whether specialization predicts *future* performance even after the bank's ex ante rating has been controlled for.
- **Empirical applications.** The paper studies five broad objects: the existence and persistence of specialization, loan performance, loan terms, aggregate bank stability/profitability, and extensions on SMEs, monitoring, liquidity provision, and deposit shocks.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper's conceptual mechanism is simple: information acquisition has increasing returns to scale. A bank learns more about an industry the more it lends in that industry, and that knowledge then makes further lending there more attractive.

This logic generates four hypotheses.

- **Hypothesis 1.** Banks with informational advantages will be underdiversified and will specialize their loan portfolios.
- **Hypothesis 2.** If specialization reflects information rather than simple constraints, loans in a bank's specialized industry should perform better. Pure concentration among constrained small lenders need not improve performance.

- **Hypothesis 3.** Because banks face risk constraints and cannot simply fill their portfolio with obviously risky loans, information-based specialization should be associated with more stable loan portfolios and fewer aggregate defaults.
- **Hypothesis 4.** Specialization has costs because banks may need to share rents with attractive borrowers through better loan terms. In ordinary times, banks may tilt more strongly toward their specialized industry when stability becomes especially valuable, for example after a hit to capital.

3.2 Measuring specialization

Let $LoanAmount_{b,s,t}$ denote the amount bank b lends in industry or sector s at time t. The paper's baseline specialization measure is

$$\text{Excess Specialization}_{b,s,t} = \frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} - \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}. \tag{1}$$

The alternative relative specialization measure is

$$\text{Relative Specialization}_{b,s,t} = \frac{\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}}}{\frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}}. \tag{2}$$

- **Interpretation of (1).** A bank is specialized if it lends more to an industry than the industry's share of total C&I lending would imply under diversification.
- **Interpretation of (2).** This expresses overinvestment as a ratio instead of a difference. A value of 2 means the bank lends twice as much to that industry as a diversified bank would.
- **Favorite industry.** The bank's top industry is the industry with the largest excess specialization.

3.3 Baseline loan-performance regression

The core regression in the paper is

$$\begin{aligned} NonPerform_{l,i,b,s,T} = \beta_0 + \beta_1 Specialization_{b,s,t} + \beta_2 X_{l,b} + \beta_3 Relationship_{i,b} + \beta_4 ShareZip_{i,z,t} \\ + \xi_{b,t} + \sigma_{s,t} + \phi_{\text{loan risk rating}} + \omega_{\text{loan purpose}} + \varepsilon_{l,i,b,s,t}, \end{aligned} \tag{3}$$

where loan l is made by bank b to borrower i in industry s at origination date t, and T indexes the future horizon over which the loan may become nonperforming.

- **Dependent variable.** Whether the loan ever becomes nonperforming after origination.
- **Main regressor.** A bank's specialization in the borrower's industry at origination.
- **Controls $X_{l,b}$.** Loan size, interest rate, maturity, collateral type, and related loan characteristics.
- **Relationship term.** Past and future bank-borrower interactions, meant to separate industry expertise from borrower-specific relationship lending.

- **ShareZip control.** A control for regional concentration in the borrower's zip code.
- **Bank-time fixed effects $\xi_{b,t}$.** Compare loans made by the same bank across industries at the same time.
- **Industry-time fixed effects $\sigma_{s,t}$.** Compare loans in the same industry and time across banks with different specialization.
- **Loan-rating fixed effects.** Help control for ex ante observables about borrower risk.

3.4 Why ratings and rates matter for interpretation

The paper wants to separate **screening** from **monitoring** as much as the data allow.

- If specialization predicts performance *without* controlling for rates and internal ratings, that could reflect better screening, better monitoring, or both.
- If specialization still predicts performance *after* controlling for origination rates and supervisory risk ratings, then part of the effect must come from information or monitoring beyond what is captured in those ex ante observables.
- This is not a perfect decomposition, but it is the paper's main way of pushing the result away from pure selection on observables.

3.5 Alternative performance specifications

The paper then estimates several variants of equation (3).

- **Horse-race regression.** Include specialization, industry portfolio share, regional concentration, and relationship lending together to see whether specialization still matters.
- **Alternative definitions.** Replace baseline excess specialization with relative specialization, four-digit specialization, portfolio shares, or a favorite-industry dummy.
- **SNC size interactions.** Interact specialization with lender-size buckets to test whether concentration among small lenders is fundamentally different from specialization among large unconstrained banks.
- **Arranger specialization.** In syndicated loans, test whether the lead arranger's specialization matters, and whether the result survives conditioning on that specialization.

3.6 Loan-terms regressions

For loan terms, the paper estimates the same basic specification but changes the dependent variable:

$$Y_{l,i,b,s,t} = \beta_0+\beta_1 Specialization_{b,s,t}+\beta_2 X_{l,b}+\beta_3 Relationship_{i,b}+\xi_{b,t}+\sigma_{s,t}+\phi_{\text{rating}}+\omega_{\text{purpose}}+\varepsilon_{l,i,b,s,t}, \tag{4}$$

with

$$Y_{l,i,b,s,t} \in \{\text{interest rate},\ \log\ \text{loan amount},\ \text{maturity},\ \text{unsecured dummy}\}.$$

The paper also estimates interactions with an unsecured-loan indicator.

- **Interpretation.** These regressions ask how specialists attract and retain borrowers they value.
- **Important caveat.** The paper is explicit that loan terms are codetermined in equilibrium, so these are correlations rather than clean causal effects.

3.7 Bank-level and deposit-shock exercises

The paper also estimates stylized collapsed regressions at the bank-time or bank-industry-time level.

$$BankOutcome_{b,t} = \alpha + \gamma\, TopIndustrySpecialization_{b,t} + \Gamma\, BankControls_{b,t} + u_{b,t}, \tag{5}$$

where outcomes include profitability, return stability, charge-offs, or the response of specialization to Tier 1 capital and risk-weighted assets.

For the COVID deposit-shock exercise, the paper relates unexpected deposit inflows to lending in specialized industries using bank-industry-time panels of the form

$$LoanGrowth_{b,s,t} = \alpha + \delta\, DepositShock_{b,t} \times Specialization_{b,s,t} + \text{controls} + u_{b,s,t}. \tag{6}$$

- **Interpretation.** These exercises show that specialization matters not only for single-loan outcomes but also for bank balance sheets and for where new funds flow after an exogenous liquidity shock.
- **Caution.** The exact implementation differs by exercise and many of these results are reported in the appendix, but the economic object is always the same: does a bank's favored industry receive systematically different treatment?

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the empirical question hard

- **Observed loans are equilibrium outcomes.** The data contain originated or held loans, not rejected applications. This makes causal interpretation difficult because the bank and borrower jointly sort into contracts.
- **Specialization may proxy for something else.** A specialized bank may simply have monopoly power in an industry, may lend where it gathers deposits, or may know specific borrowers rather than industries.
- **Selection of safe borrowers.** Better performance could just mean specialists cherry-pick safer firms.
- **Industry measurement.** Borrower industries in Y-14 are reported by banks and may be noisy.
- **Constraint story.** A small or constrained lender can look fully concentrated without possessing any real informational edge.

4.2 What the paper does to make the results credible

This is not an IV paper. Instead, credibility comes from **saturated comparisons**, **alternative controls**, and **heterogeneity tests**.

- **Bank-time and industry-time fixed effects.** These force the comparison to be within the same bank at the same time and within the same industry at the same time.
- **Loan-level observables.** The paper controls for size, rate, maturity, purpose, and collateral type.
- **Internal risk ratings.** If specialization still predicts performance after conditioning on origination ratings, the result is harder to explain by pure observables-based selection.
- **Relationship lending and geography.** The paper adds bank-borrower interaction measures, zip-level lending concentration, deposit concentration, and excess regional lending.
- **Industry capture controls.** These help separate information-based specialization from rent extraction by banks that dominate a sector.
- **Alternative specialization definitions.** Two-digit versus four-digit, excess versus relative, portfolio shares, and favorite-industry dummies all give the same sign.
- **Alternative industry coding.** Results survive when industries are measured using SIC codes in SNC or primary industries from Compustat in the merged subsample.
- **Large-versus-small-lender evidence.** In SNC, specialization predicts worse performance for small constrained lenders but better performance for large unconstrained lenders. This is exactly what the paper's information-versus-constraint story predicts.
- **Additional robustness.** The paper reports firm fixed-effects specifications for the subset of firms borrowing from multiple banks in the same period, dynamic-panel regressions, and Cox hazard models; the main pattern remains.

4.3 How this relates to the literature

- **Specialization versus diversification.** The paper joins the debate on whether concentrated bank portfolios are stabilizing because of information or destabilizing because of lack of diversification.
- **Beyond relationship lending.** It argues that repeated lending to an *industry*, not just to a borrower, creates information capital.
- **Related work.** The closest paper is Paravisini, Rappoport, and Schnabl (2020), who study specialization across export markets in Peru. Blickle, Parlatore, and Saunders push this logic to industry specialization by large U.S. banks.
- **Banking interpretation.** The paper emphasizes banks as information-producing intermediaries, not just liquidity transformers.

5 Tables and figures

5.1 Table I: Summary statistics of key variables

Read:

- The main Y-14 term-loan sample is large and economically meaningful: 296,951 loans, average interest rate 3.80%, average remaining maturity 24.5 quarters, and a 4% lifetime nonperformance rate.

- Loans made in a bank's favorite industry are better on multiple raw dimensions: they are larger (log amount 8.50 versus 8.43), cheaper (3.61% versus 3.83%), less often unsecured (13% versus 19%), and less likely to become nonperforming (3% versus 4%).

- The average bank-firm pair is observed repeatedly, which is why the paper can control for relationship lending.

- Table I is the paper's first indication that specialization is associated with both different loan terms and better subsequent performance.

Table I

Summary Statistics of Key Variables

This table presents summary statistics for the loans in our sample. We count each bank-loan combination only once, on the date when it is first observed in our data (this may be a different date than the loan's first origination date for some early loans). Log size is based on the natural logarithm of the committed exposure, scaled by 1000 USD. The interest rate is the unadjusted cost of the loan, measured in percent. Maturity is measured in quarters remaining. Unsecured is a dummy that takes the value of 1 if the loan is not secured with any type of collateral. "Nonperforming" is a dummy that takes the value of 1 if the loan falls in arrears, has negative maturity, or is otherwise in default. Industry capture, the share of all lending to that industry in the Y-14 data, measures the degree to which a given bank accounts for all lending to the two-digit industry of the borrower. Finally, bank-firm interactions are the number of times a given borrower and lender ever interacted in the Y-14 data. The mean values of each variable are split by whether a loan is made in a lender's favorite (i.e., "top") industry.

	N	Mean	*SD*	Top-Industry	Other-Industry	Diff
Log Amount	296,951	8.42	1.16	8.50	8.43	0.07***
Interest Rate	296,951	3.80	1.70	3.61	3.83	−0.21***
Maturity Remaining	296,951	24.51	21.05	19.84	18.40	1.4***
Unsecured	296,951	0.18	0.38	0.13	0.19	−0.06***
Loan Becomes nonperforming	296,951	0.04	0.19	0.03	0.04	−0.01***
Industry Capture	296,951	0.09	0.098	0.086	0.073	0.013***
Bank-Firm Interactions	296,951	9.32	37.2	6.93	9.69	−2.7***

that takes the value of 1 if a loan is flagged as nonaccruing or in default by the reporting bank, if the loan is more than 89 days in arrears, or if the loan has any negative maturity because some amount of the principal or interest has not been repayed at loan-end. Just over 4% of term loans ever become nonperforming during their time in the sample.

The degree to which an industry has been captured by a single bank is around 9% in our sample, which means that the average bank accounts for 9% of C&I lending to a certain two-digit industry, as defined within the Y-14 sample. The average number of interactions between a bank and a borrower is nine. This figure naturally rises toward the end of our sample and is driven by larger institutions. The regressions below use both past bank-borrower interactions (for which the average is 4.2) and total interactions to capture a "relationship" between a bank-borrower pair.

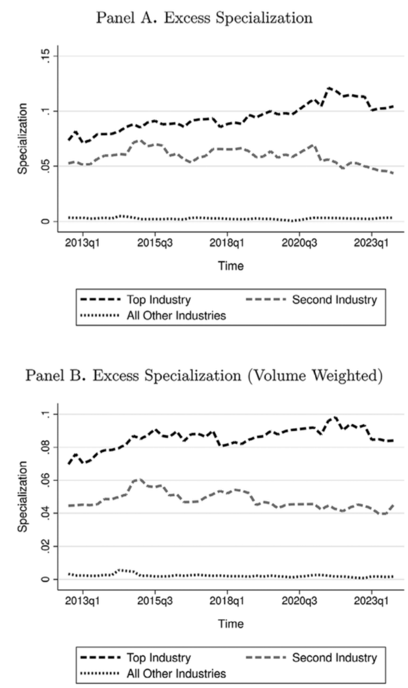

Figure 1. Excess specialization. This figure shows the degree to which banks in our data are "overinvested" in their favorite, second favorite, and all other industries. A perfectly diversified bank is one that is invested in accordance with industry size, with larger industries (relative to all C&I lending) receiving proportionately more loans. We use excess specialization, defined as $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} - \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$; for term loans in our sample. A favorite industry is one in which a bank is the most overinvested. Panel A depicts unweighted calculations of average specialization, while Panel B weights our specialization measures according to committed loan amounts before averaging.

5.2 Figure 1: Excess specialization over time

Read:

- The figure is the paper's main descriptive fact. The average bank overinvests heavily in one or two industries and not in the rest.

- In unweighted terms, the excess share in the top industry is usually around 0.07 to 0.12, the second-favorite industry around 0.05 to 0.07, and all other industries are near zero.

- The volume-weighted version looks similar, so the pattern is not driven only by small banks.

- Specialization is persistent, not a transient artifact of a few large loans maturing.

5.3 Table II: Summary statistics of specialization

Read:

- At the two-digit level, the average bank's top industry has excess specialization of **0.09**, relative specialization of **3.56**, and a portfolio share of **18%**; all other industries have excess specialization of only about **0.01** and portfolio share of **5%**.
- At the four-digit level, the top sector still stands out strongly: excess specialization **0.04**, relative specialization **5.92**, and portfolio share **7%**.
- The gap between top-industry and other-industry lending is too large to be explained by mild portfolio tilts.
- The table quantifies the idea that banks have one clear favorite industry, maybe a second one, and much weaker exposure elsewhere.

Table II

Summary Statistics of Specialization

This table presents summary statistics for various specialization measures at the two-digit industry and four-digit sector levels. Specialization is defined as the degree to which a bank is overinvested in an industry, relative to a perfectly diversified portfolio. A diversified portfolio is one based solely on the size of an industry relative to all C&I lending. We show relative specialization (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} / \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$), excess specialization (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} - \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$), and portfolio share (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}}$). We split data by whether an industry/sector is a bank's most preferred "top" industry (as measured by the respective specialization measure) or not.

		Top Industry				All Other Industries			
	Specialization Type	Mean	*SD*	25-pct	75-pct	Mean	*SD*	25-pct	75-pct
Two Digit	Relative Specialization	3.56	1.30	2.37	5.12	1.26	0.71	0.80	1.56
	Excess Specialization	0.09	0.05	0.07	0.14	0.01	0.03	−0.01	0.02
	Portfolio Share	0.18	0.14	0.11	0.20	0.05	0.04	0.03	0.07
Four Digit	Relative Specialization	5.92	1.89	6.71	6.96	1.97	1.71	0.79	4.59
	Excess Specialization	0.04	0.01	0.03	0.05	0.01	0.02	−0.00	0.01
	Portfolio Share	0.07	0.05	0.06	0.07	0.03	0.08	0.00	0.02

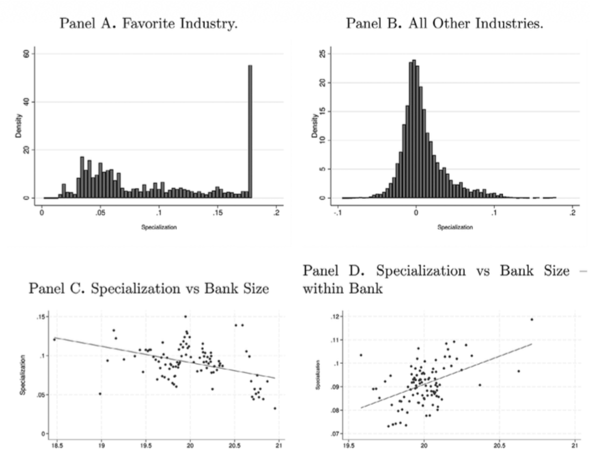

Figure 2. Variance of excess specialization and bank size. This figure plots the distribution of excess specialization for the banks in our data in Panels A and B. Excess specialization is measured as the degree to which a bank is overinvested in an industry. A perfectly diversified bank is one that is invested by industry size, so that excess specialization is given by $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} - \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$. We split our sample into a bank's favored industry (Panel A) and all other industries (Panel B). Panels C and D relate specialization to bank size using bin scatters, where each bin represents at least five observations. Panel C accounts for bank category type and Panel D for bank fixed effects.

5.4 Figure 2: Distribution of specialization and bank size

Read:

- Panel A shows substantial heterogeneity in the degree of favorite-industry specialization: some banks are only mildly tilted, while others are very concentrated.
- Panel B shows that specialization in nonfavored industries is centered much closer to zero.
- Panel C shows that larger banks are less specialized *across banks*, but still have preferred industries.

- Panel D shows that *within a bank*, growth is associated with more specialization, consistent with banks "doubling down" in their favored industry.

5.5 Figure 3: Raw loan-term and performance differences

Read:

- Specialized-industry loans are systematically larger than nonspecialized-industry loans, both in raw comparisons and after detailed controls.
- Specialized-industry loans also carry lower interest rates and lower nonperformance rates over almost the entire sample window.
- Panel D shows a fairly linear negative relation between specialization and future nonperformance: the more specialized the lender, the safer the loan looks ex post.
- The figure is useful because it lets you see that the main regression results are already visible in raw time-series comparisons.

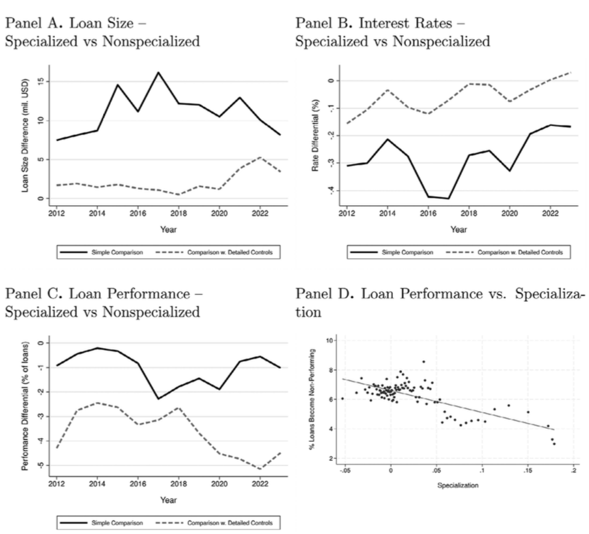

Figure 3. Excess specialization, loan terms, and loan performance. This figure shows the difference between the size (Panel A), the rates charged (Panel B), and the likelihood of becoming nonperforming (Panel C) for loans granted by banks to their favorite industry vs. all other industries. Solid lines are raw differences and dashed lines account for loan rating, purpose, and industry*time. A bank's favorite industry is the one in which it is most overinvested (i.e., where $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} - \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$ is largest for term loans in our sample). Panel D relates the propensity of loans to become nonperforming to bank specialization in the borrower's industry.

Table III

Specialization and Loan Performance

This table presents the coefficients of interest for:

$$NonPerformance_{l,i,b,s,T} = \beta_0 + \beta_l Specialization_{b,s,t} + \beta_2 \mathbf{X}_{l,b} + \beta_3 Relationship_{i,b} + \beta_4 ShareZip_{i,z,t} \xi_{b,t} + \sigma_{s,t} + \phi_{loanriskrating} + \omega_{loanpurpose} + \epsilon_{l,i,b,s,t},$$

which regresses whether loan l to firm i in quarter t operating in sector/industry s and located in zip code z, which is made by bank b, ever becomes nonperforming in future periods on bank b's specialization in industry s. "Nonperforming" is a dummy that takes the value of 1 if the loan ever falls in arrears, has negative maturity, or is otherwise in default after it is first observed in our data. Specialization is defined as the degree to which a bank is overinvested in an industry, relative to a perfectly diversified portfolio. A diversified portfolio is one based solely on the size of an industry relative to all C&I lending. We define "excess" (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} - \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$)) specialization at the two-digit industry level. Columns include additional controls and fixed effects as specified. Column (1) includes time-invariant bank fixed effects as well as purpose and quarter*year. Columns (2) to (5) additionally include industry*time and bank*time fixed effects. We account for the interest rate paid in columns (3) to (5), the loan's rating at first observation in columns (4) to (5), and the size of the loan in columns (4) to (5). We include dummies for different collateral types pledged in column (5), including real estate, marketable securities, accounts receivable, fixed assets, blanket lien, and "other." Our data focus on the first observation of a loan and contains only term loans. Standard errors are clustered at the bank-industry-year level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Loan Ever Becomes Nonperforming				
	(1)	(2)	(3)	(4)	(5)
Excess Specialization	−0.156***	−0.139***	−0.121***	−0.091***	−0.098***
	[0.016]	[0.014]	[0.014]	[0.013]	[0.013]
Interest rate			0.014***	0.007***	0.007***
			[0.000]	[0.000]	[0.000]
Log loan amount			0.000	0.001***	0.001**
			[0.000]	[0.000]	[0.000]
Share of Portfolio in ZIP					−0.013*
					[0.007]
Number of past interactions (relationship)					−0.000
					[0.000]
Future Interactions (relationship)					−0.001***
					[0.000]
General Fixed Effects	Purpose, Time				
Specific Fixed Effects	Bank	Industry*Time, Bank*Time			
Loan Rating Fixed Effects	No	No	No	Yes	Yes
Collateral Fixed Effects	No	No	No	No	Yes
Mean of dependent variable	0.04	0.04	0.04	0.04	0.04
R^2	0.011	0.037	0.046	0.14	0.14
N	298,043	298,043	298,043	296,951	296,951

5.6 Table III: Specialization and loan performance

Read:

- This is the core table. The coefficient on excess specialization is **-0.156** with basic fixed effects and remains **-0.098** after adding bank-time and industry-time effects, interest rates, loan size, internal ratings, collateral, relationship measures, and regional concentration.
- In the paper's own translation, moving from a bank's average industry to its favorite industry lowers the probability that a loan becomes nonperforming by about **1.3 percentage points** in the simple specification; the effect remains economically meaningful after the full set of controls.
- Conditioning on origination ratings shrinks the coefficient but does not kill it, which is consistent with specialists doing more than just choosing loans that obviously look safe on hard information.
- Future bank-borrower interactions also predict better performance, but industry specialization survives even after controlling for that relationship channel.

5.7 Table IV: Alternative concentration measures and the horse race

Read:

- When the paper includes industry specialization, industry portfolio share, zip-code concentration, and relationship lending in the same regression, specialization still matters.
- In the horse race, a one-standard-deviation increase in specialization lowers nonperformance by nearly **1 percentage point**, even in a saturated specification.
- Regional concentration and relationship lending matter too, but they do not subsume industry specialization.
- This table is important because it shows that specialization is not just a relabeling of geography, size, or repeated lending to the same firm.

5.8 Table V: Different specialization definitions

Read:

- The main result is robust to basically every specialization measure the paper tries: two-digit excess, four-digit excess, two-digit relative, four-digit relative, portfolio-share measures, and a favorite-industry dummy.
- With the full controls, the coefficient on two-digit excess specialization is **-0.098**; on four-digit excess specialization it is **-0.065**; and the favorite-industry dummy is **-0.007**.
- The four-digit effects are smaller, which makes sense because a bank can have several closely related favorite four-digit sectors inside one broad two-digit industry.
- The message is that the result is not an artifact of one scaling choice or one tail treatment.

5.9 Table VI: Specialization and loan terms

Read:

Table IV

Alternative Specialization Measures and Loan Performance

This table presents the coefficients of interest for:

$$NonPerformance_{l,i,b,s,T} = \beta_0 + \beta_l Specialization_{b,s,t} + \beta_2 \mathbf{X}_{l,b} + \xi_{b,t} + \sigma_{s,t} + \phi_{loanriskrating} + \omega_{loanpurpose} + \epsilon_{l,i,b,s,t},$$

which regresses whether loan l to firm i in quarter t operating in sector/industry s and located in zip code z, which is made by bank b, ever becomes nonperforming in future periods on bank b's specialization in industry s. "Nonperforming" is a dummy that takes the value of 1 if the loan ever falls in arrears, has negative maturity, or is otherwise in default after it is first observed in our data. Specialization is defined in several different ways. First, we use the degree to which a bank is overinvested in an industry, relative to a perfectly diversified portfolio. A diversified portfolio is one based solely on the size of an industry relative to all C&I lending. We use excess specialization (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} - \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$)) at the two-digit industry level. Second, we use the share of a bank's portfolio invested in a single industry (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}}$). Third, we use the share of a bank's portfolio invested in a single zip code (i.e., $\frac{LoanAmount_{b,z,t}}{\sum_s LoanAmount_{b,z,t}}$). Finally, we use the relationship between a bank and a borrower, measured as the number of interactions the two have over our entire data set. Column (6) makes use of standardized coefficients. All columns contain bank*time, collateral type (including real estate, marketable securities, accounts receivable, fixed assets, blanket lien, and "other"), and loan purpose fixed effects. We additionally account for the interest rate paid, the loan's rating at first observation, and the size of the loan. Columns (1) to (4) include industry*time fixed effects. Our data focus on the first observation of a loan and contain only term loans. Standard errors are clustered at the bank-industry-year level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

	Loan ever becomes non-performing					
	(1)	(2)	(3)	(4)	(5)	(6)
Excess Specialization	−0.097***				−0.096***	−0.007***
	[0.013]				[0.026]	[0.002]
Share of Portfolio in industry		−0.089***			−0.020	−0.002
		[0.012]			[0.017]	[0.004]
Share of Portfolio in ZIP			−0.015**		−0.012	−0.001
			[0.007]		[0.007]	[0.00]
Borrower Relationship				−0.001***	−0.001***	−0.003***
				[0.000]	[0.000]	[0.001]
Fixed Effects	Bank*Time, Loan Purpose, Loan Rating					
Industry Time FE	Yes	Yes	Yes	Yes	No	No
Controls	Loan Rate, Size, Maturity, Bank Industry Capture, Collateral					
Standardized Coefficients	No	No	No	No	No	Yes
Mean of dependent variable	0.05	0.04	0.04	0.04	0.04	0.04
R^2	0.16	0.16	0.16	0.16	0.17	0.17
N	296,951	296,951	296,951	296,951	296,951	296,951

Table V

Loan Performance—Different Possible Specialization Measures

Each coefficient in this table corresponds to a stand-alone regression. We vary our measure of specialization in:

$$Y_{l,i,b,s,T} = \beta_0 + \beta_l Specialization_{b,s,t} + \beta_2 \mathbf{X}_{l,b} + \beta_3 Relationship_{i,b} + \xi_{b,t} + \sigma_{s,t} + \phi_{loanriskrating} + \omega_{loanpurpose} + \epsilon_{l,i,b,s,t},$$

which regresses whether loan l to firm i in quarter t by bank b in sector/industry s ever becomes nonperforming in future periods on the lending bank b's specialization in industry s. "Nonperforming" is a dummy that takes the value of 1 if the loan falls in arrears or is otherwise in default. Specialization is defined as the degree to which a bank is overinvested in an industry, relative to a perfectly diversified portfolio. A diversified portfolio is one based solely on the size of an industry relative to all C&I lending. We use both relative (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} / \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$) and excess (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} - \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$) specialization at the two-digit and four-digit level. Further, we use the degree to which a bank has invested in a portfolio at the two- and four-digit levels as a further measure of specialization (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}}$) and whether a two-digit industry is a bank's favorite industry (i.e., a dummy). All specifications include bank*time, industry*time, and loan purpose fixed effects as well as loan size, bank-borrower relationship measures such as past and future bank-borrower interactions and the share of bank b's portfolio in borrower i's zip code. Columns (2) to (3) add collateral and interest rate controls and column (3) adds loan rating at origination. Our sample includes only term loans when first observed in the sample. Standard errors are clustered at the bank-industry-year level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

		Loan Ever Becomes Nonperforming		
		(1)	(2)	(3)
Excess Specialization	2-Digit Industry	−0.121***	−0.091***	−0.098***
		[0.014]	[0.013]	[0.013]
	4-Digit Industry	−0.098***	−0.049**	−0.065***
		[0.026]	[0.025]	[0.024]
Relative Specialization	2-Digit Industry	−0.005***	−0.004***	−0.005***
		[0.001]	[0.001]	[0.001]
	4-Digit Industry	−0.001**	−0.001*	−0.000*
		[0.000]	[0.000]	[0.000]
Portfolio Share	2-Digit Industry	−0.106***	−0.076***	−0.081***
		[0.012]	[0.012]	[0.012]
	4-Digit Industry	−0.070***	−0.038***	−0.046***
		[0.013]	[0.013]	[0.013]
Top Industry Dummy		−0.009***	−0.006***	−0.007***
		[0.002]	[0.001]	[0.001]
Baseline FE:		Bank*Time, Ind.*Time, Purpose		
Loan Rating at First Obs.		**No**	**No**	**Yes**
Interest Rate and Collateral Controls		**No**	**Yes**	**Yes**
Other Loan and Bank Controls		Yes	Yes	Yes
Loan Purpose and Type FE		Yes	Yes	Yes
N		298,043	296,951	296,951

Table VI

Specialization and Loan Terms

This table relates loan terms to a bank's specialization in that industry and a series of controls. We estimate the regression

$$Y_{l,i,b,s,T} = \beta_0 + \beta_l Specialization_{b,s,t} + \beta_2 \mathbf{X}_{l,b} + \beta_3 Relationship_{i,b} + \xi_{b,t} + \sigma_{s,t} + \phi_{loanriskrating} + \omega_{loanpurpose} + \epsilon_{l,i,b,s,t},$$

which regresses the terms of loan l to firm i in quarter t by bank b in sector/industry s on the lending bank b's specialization in industry s. Specialization is defined as the degree to which a bank is overinvested in an industry, relative to a perfectly diversified portfolio. A diversified portfolio is one based solely on the size of an industry relative to all C&I lending. We use excess specialization (i.e., $\frac{LoanAmount_{b,s,t}}{\sum_s LoanAmount_{b,s,t}} - \frac{LoanAmount_{s,t}}{\sum_s LoanAmount_{s,t}}$) at the two-digit level. All specifications include bank*time, industry*time, and loan purpose fixed effects. We include loan terms that are not the dependent variable as controls including loan size, loan rate, and loan maturity. We include dummies for different collateral types pledged, including real estate, marketable securities, accounts receivable, fixed assets, blanket lien, and "other" in columns (1) to (3). Panel A makes use of the baseline specification. Panel B includes an interaction term for our variable of interest that takes the value of 1 if the loan is not secured by collateral. We exclude collateral fixed effects from these regressions. Our sample includes only term loans when first observed in the sample. Standard errors are clustered at the bank-industry-year level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Panel A

	Interest Rate (1)	Log Loan Amount (2)	Maturity Remaining (3)	Unsecured (4)
Excess Specialization	−0.569***	1.242***	3.803*	−0.044
	[0.134]	[0.114]	[2.008]	[0.043]
Fixed Effects	Bank*Time, Industry*Time, Loan Purpose, Loan Rating			
Controls	Loan Rate, Size, Maturity, Bank Industry Capture, Collateral			
Mean of dependent variable	3.6	8.2	22	0.14
R^2	0.36	0.25	0.28	0.38
N	296,951	296,951	296,951	296,951

Panel B

	Interest Rate (1)	Log Loan Amount (2)	Maturity Remaining (3)	— (4)
Excess Specialization	−0.193	1.220***	3.690*	–
	[0.154]	[0.107]	[2.047]	–
Int.: Unsecured * Specialization	−1.583***	0.294	15.267***	–
	[0.366]	[0.320]	[5.665]	–
Fixed Effects	Bank*Time, Industry*Time, Loan Purpose, Loan Rating			
Controls	Loan Rate, Size, Maturity, Bank Industry Capture, Collateral			
Mean of dependent variable	3.6	8.2	22	–
R^2	0.42	0.26	0.28	–
N	296,951	296,951	296,951	–

- Specialization is associated with more attractive loan terms: the coefficient on excess specialization is **-0.569** for interest rates, **1.242** for log loan amount, and **3.803** for remaining maturity in the baseline controlled specification.
- The unsecured-loan interaction shows that the rate and maturity effects are especially strong for unsecured loans. In other words, industry knowledge matters most when collateral is weakest.
- The table supports the paper's rent-sharing story: specialized banks seem willing to give up some raw profitability to attract borrowers they view as especially valuable.
- These are equilibrium correlations, not causal estimates, but they fit the theory of information-based specialization remarkably well.

5.10 Aggregate results and extensions from Sections VII–VIII

Read:

- At the bank level, higher top-industry specialization is associated with somewhat **lower profitability** but more **stable returns** and fewer charge-offs. The paper interprets this as a trade-off between rent extraction and stability.
- Banks with lower relative Tier 1 capital or higher risk-weighted assets refocus more strongly on their preferred industry, consistent with specialization being used as a stabilizing strategy.
- Specialized banks are more likely to lend to SMEs and opaque borrowers, to actively monitor loans, and to provide liquidity to distressed borrowers in their specialized industries.
- During COVID, deposits at sample banks rose by about **30% in a single quarter**, and to the extent those funds flowed into C&I lending they were directed disproportionately toward the banks' preferred industries. Firms tied to more specialized lenders saw stronger growth and earnings in that period.

6 Short summary

- Large U.S. banks do not hold broad "market" loan portfolios; they specialize in one or two industries.
- This specialization is most consistent with industry-specific information, because it predicts better loan performance even after rich controls and because the effect is strongest for large unconstrained lenders.
- Specialists appear to share some of the gains from information with borrowers through bigger, cheaper, and longer loans, especially when information problems are severe.
- The cost is lower raw profitability and greater exposure to downturns in the preferred industry, but the benefit is greater loan stability and better performance in ordinary times.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that even the largest U.S. banks are not close to fully diversified in C&I lending. They choose and maintain industry specialization.
- That specialization is closely linked to better ex post loan outcomes in the preferred industry.
- The evidence is most naturally interpreted as banks acquiring and deploying industry-specific information that improves screening and monitoring.
- The paper also shows that specialization affects contract design, bank risk, and the way exogenous funding shocks get transmitted to firms.

7.2 Economic intuition

A bank that repeatedly lends in one industry learns how to evaluate business models, collateral, and distress in that industry. That knowledge helps it distinguish good from bad borrowers more accurately and may also help it intervene more effectively when trouble arrives. But specialization is not free. To win the best borrowers, the bank may need to offer better terms, and concentrating too much in one industry exposes it to sector-specific downturns. The observed data line up with exactly this trade-off: specialists make safer loans and have more stable portfolios, but they also seem willing to sacrifice some profitability and take on industry concentration risk.

7.3 Takeaway

This paper should be read as a statement that bank credit supply is **industry-shaped**, not generic. A dollar of deposits flowing into one bank is not the same as a dollar flowing into another bank, because different banks channel funds toward different industries where they possess different amounts of information. For work on bank lending, credit reallocation, banking stress, industrial policy, or the real effects of deposit flows, that is the paper's central lesson.

Equally important is the paper's caution about interpretation. The authors do not observe rejected applications and therefore do not claim a clean causal design. Instead, they show that the same specialization-performance pattern survives a large number of fixed effects, controls, alternative definitions, and auxiliary data sets. That combination makes the paper especially useful as a benchmark for future work on bank information, bank heterogeneity, and the real effects of credit reallocation.